

# PAPER\_851: Kozima LENR Neutron Drop — Density-Scaled 8-System Batch with PSR J0030+0451

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## Abstract

Kozima's neutron drop model (PMC8141838, 2021) is extended within UQFF through three complementary formulations of the neutron absorption cross-section: static ( $\sigma_n = 10^{-4} \text{ m}^2$ ), frequency-dependent ( $\sigma_n(\omega)$  with Gaussian resonance at  $\omega_{\text{LENR}}$ ), and density-scaled ( $\sigma_n(\rho) = \sigma_0 \times \rho/\rho_0$ ). Eight systems are recalculated with  $F_{\text{neutron}}$ , and the density-scaled model predicts  $F_{\text{neutron}} \sim 10^{45} \text{ N}$  for neutron star PSR J0030+0451 ( $\rho \sim 10^{17} \text{ kg/m}^3$ ).

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## 1. Kozima Neutron Drop Model

Source: Kozima H. "Cold Fusion: A Hypothesis on the Reaction Process in a Lattice" (PMC8141838, 2021)

Central idea: neutrons form stable "drops" within metal lattices at specific phonon frequencies, enabling nuclear transmutation without Coulomb barrier penetration.

$F_{\text{neutron}} = k_{\text{neutron}} * \sigma_n$   
 $k_{\text{neutron}} = 10^{10} \text{ N/m}^2$   
 $\sigma_n = 10^{-4} \text{ m}^2$  (general)  
 $F_{\text{neutron}} = 10^6 \text{ N}$  (static)

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## 2. Three Formulations

### 2.1 Static Model

$F_{\text{neutron}} = k_{\text{neutron}} * \sigma_0 = 10^{10} * 10^{-4} = 10^6 \text{ N}$

Applies to: lab-scale LENR with ambient neutron flux

### 2.2 Frequency-Dependent Model

$\sigma_n(\omega) = \sigma_0 * (\omega/\omega_{\text{LENR}})^2 * \exp(-(\omega - \omega_{\text{LENR}})^2 / (2 * \Delta\omega^2))$

$\omega_{\text{LENR}} = 2 * \pi * 1.25 \text{e}12 = 7.854 \text{e}12 \text{ rad/s}$

$\Delta\omega = 2 * \pi * 0.05 \text{e}12 = 3.14 \text{e}11 \text{ rad/s}$  (bandwidth)

At resonance ( $\omega = \omega_{\text{LENR}}$ ):

$\sigma_n = \sigma_0 * 1 * \exp(0) = \sigma_0 = 10^{-4} \text{ m}^2$

Off-resonance rapidly suppressed by Gaussian envelope.

Dynamic form:

$F_{\text{neutron}}(t) = k_{\text{neutron}} * \sigma_n(\omega_{\text{eff}}) * (1 + \alpha * \cos(\omega_{\text{act}} * t))$   
 $\alpha = 0.1$ ,  $\omega_{\text{eff}} = \omega_{\text{act}} + n * \omega_{\text{LENR}}$  ( $n \sim 4.17e9$ )

## 2.3 Density-Scaled Model

$\sigma_n(\rho) = \sigma_0 * (\rho / \rho_0)$   
 $\rho_0 = 10^{-22} \text{ kg/m}^3$  (reference vacuum density)

| Environment  | $\rho$ (kg/m <sup>3</sup> ) | $\sigma_n$ (m <sup>2</sup> ) | $F_{\text{neutron}}$ (N) |
|--------------|-----------------------------|------------------------------|--------------------------|
| Vacuum       | $10^{-22}$                  | $10^{-4}$                    | $10^6$                   |
| Laboratory   | $10^{-10}$                  | $10^8$                       | $10^{18}$                |
| Solid        | $10^3$                      | $10^{21}$                    | $10^{31}$                |
| White dwarf  | $10^9$                      | $10^{27}$                    | $10^{37}$                |
| Neutron star | $10^{17}$                   | $10^{35}$                    | $10^{45}$                |

## 3. 8-System Recalculation with $F_{\text{neutron}}$

All 8 sonification systems recalculated:

$F_{\text{neutron}}(\text{static}) = 10^6 \text{ N}$  for all astrophysical systems (no density data)

For neutron-star-density systems:

PSR J0030+0451:  $\rho \sim 10^{17} \text{ kg/m}^3$

$F_{\text{neutron}}(\text{density}) = 10^{10} * 10^{-4} * (10^{17} / 10^{-22}) = 10^{45} \text{ N}$

This exceeds  $F_{\text{LENR}}$  ( $6.17e37 \text{ N}$ ) by 8 orders of magnitude at neutron star densities.

## 4. PSR J0030+0451

Millisecond pulsar:  $P = 4.87 \text{ ms}$

$M = 1.44 \pm 0.15 M_{\text{sun}}$  (NICER 2019)

$R = 13.02 \pm 0.87 / -0.84 \text{ km}$  (NICER 2019)

$\rho_{\text{avg}} \sim 10^{17} \text{ kg/m}^3$

Surface gravity:  $g = GM/R^2 = 6.674e-11 * 2.863e30 / (1.3e4)^2$   
 $= 1.13e12 \text{ m/s}^2$

NICER hotspot analysis: two hotspot regions on surface

X-ray pulse profile constrains  $M/R$  ratio

$F_{\text{neutron}}(\text{PSR J0030+0451}) \sim 10^{45} \text{ N}$

This would make  $F_{\text{neutron}}$  the DOMINANT term for neutron stars, exceeding  $F_{\text{LENR}}$  by  $\sim 10^8$ .

## Conclusion

Kozima's neutron drop model provides three complementary  $F_{\text{neutron}}$  formulations for UQFF. The density-scaled model predicts  $F_{\text{neutron}} \sim 10^{45} \text{ N}$  at neutron star densities, making it the dominant  $F_{\text{U\_Bi\_i}}$  term for compact objects. PSR J0030+0451 (NICER target) provides a direct observational test of this prediction.

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